

Scientists See Tiny Particles Make Ripples in the Universe's First 'Soup'

From: ScienceDaily Top Technology Mined: 2026-10-07 5 min read

Right after the universe began, it was trillions of degrees hot. Tiny **particles** called **quarks** and gluons zipped around at nearly the speed of light in a state of matter called quark-gluon **plasma**. It lasted only a few millionths of a second. Then it cooled, and the **particles** joined to form protons, neutrons and other building blocks of everything around us.

Today, physicists can make tiny amounts of this **plasma** at CERN's Large Hadron Collider in Switzerland. They smash heavy atoms, such as lead, together at nearly the speed of light. The **plasma** droplets last less than a quadrillionth of a second.

Now a team led by physicists from MIT has found clear evidence that **quarks** leave **wakes** as they move through the **plasma**, like a duck making ripples on a pond. That means the **plasma** acts like one connected liquid, with waves, splashes and swirls, instead of a crowd of separate bouncing **particles**.

"It has been a long debate in our field, on whether the **plasma** should respond to a quark," said MIT physics professor Yen-Jie Lee. "Now we see the **plasma** is incredibly dense, such that it is able to slow down a quark, and produces splashes and swirls like a liquid."

Scientists think quark-gluon **plasma** was the universe's first liquid, and the hottest liquid ever known. It flows so smoothly, with almost no **friction**, that some call it a near-"perfect" liquid.

Earlier searches were tricky because **quarks** often appear in pairs moving in opposite directions, and one can hide the other's wake. The team developed a new method to spot the **wakes**.

By measuring how big the **wakes** get and how fast they fade, scientists hope to learn more about the universe's first moments. The findings appear in the journal Physics Letters B.

Word Treasure

quarks -- Tiny particles that join together to make protons and neutrons.

plasma -- A very hot state of matter in which particles are not bound together the way they are in ordinary matter.

particle -- An extremely small piece of matter.

wakes -- Trails of waves or ripples left behind by something moving through a liquid.

friction -- A force that slows down things that rub or move against each other.

Background you need

All ordinary matter is made of atoms, and the protons and neutrons inside atoms are made of even smaller particles called quarks, held together by gluons.

Right after the Big Bang, the universe was so hot that quarks and gluons could not stick together and instead formed a hot plasma.

CERN's Large Hadron Collider is a huge ring-shaped machine that speeds up particles and smashes them together so scientists can study what comes out.

By recreating this plasma for tiny fractions of a second, physicists can learn how the early universe behaved.

Break it down

LEAD: The universe began as a trillions-of-degrees soup of quarks and gluons

KEY EVENT: Physicists find clear evidence that quarks leave wakes in quark-gluon plasma

+ - WHO: A team led by MIT physicists, including Yen-Jie Lee

+ - WHERE: CERN's Large Hadron Collider in Switzerland

+ - HOW: Smashing heavy atoms such as lead at nearly light speed

EVIDENCE: Wakes, splashes and swirls show the plasma acts like one liquid

CHALLENGE: Paired quarks hid each other's wakes, so the team built a new method

NEXT STEP: Measuring wake size and fading to learn about the universe's first moments

Why it matters

Studying this plasma helps scientists understand how the matter that makes up everything around us first formed. It also tests long-debated ideas about how this early matter behaves.

Test what you remember

1. What is quark-gluon plasma?

- ☐ (A) A new type of battery
- ☐ (B) A frozen gas on Mars
- ☐ (C) A kind of rocket fuel
- ☐ (D) Super-hot matter of quarks and gluons from the universe's first moments

2. Where do physicists make tiny amounts of this plasma?

- ☐ (A) On the International Space Station
- ☐ (B) At NASA in Florida
- ☐ (C) At the bottom of the ocean
- ☐ (D) At CERN's Large Hadron Collider in Switzerland

3. What did the team find quarks leave behind as they move through the plasma?

- ☐ (A) Sound waves only
- ☐ (B) Nothing at all
- ☐ (C) Ice crystals
- ☐ (D) Wakes, like a duck on a pond

4. What does this discovery show about the plasma?

- ☐ (A) It is made of water
- ☐ (B) It is a solid
- ☐ (C) It acts like one connected liquid
- ☐ (D) It is cold

5. Why were quark wakes hard to see before?

- ☐ (A) Quarks often come in pairs moving in opposite directions, and one can hide the other's wake
- ☐ (B) The plasma lasts for years
- ☐ (C) Quarks are too large
- ☐ (D) The machines were turned off

6. Where were the findings published?

- ☐ (A) Nature
- ☐ (B) Physics Letters B
- ☐ (C) Royal Society Open Science
- ☐ (D) Science Advances

Answer key (try the quiz first!): 1: D 2: D 3: D 4: C 5: A 6: B

1. Yen-Jie Lee, MIT physics professor: Says the long debate is answered: the plasma is dense enough to slow a quark and splashes and swirls like a liquid.